SECTION 1 Time—30 minutes 45 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. NUTRITIOUS: (A) bland (B) dietetic (C) crude (D) unusual (E) unwholesome
2. WARP: (A) remove (B) straighten (C) lengthen (D) choke (E) pierce
3. CLEAR-CUT: (A) unique (B) useless (C) undefined (D) disagreeable (E) permanent
4. VENTURESOME: (A) delinquent (B) busy (C) cautious (D) fictitious (E) profitable
5. SEVER: (A) choose (B) distribute (C) simplify (D) connect (E) report
6. WAVER: (A) fail to acknowledge (B) speak quietly (C) seem indifferent (D) pretend to believe (E) stand firm
7. HERALDED: (A) unaccompanied (B) unannounced (C) uncommitted (D) forestalled (E) foregone
8. PERPETUAL: (A) inconstant (B) sluggish (C) improbable (D) illogical (E) miniature
9. SUBMISSIVE: (A) misguided (B) suspect (C) rebellious (D) happy (E) talkative
10. PROFUSION: (A) impurity (B) division (C) originality (D) brittleness (E) scarcity
11. EUPHEMISM: (A) tasteful design (B) angry argument (C) difficult decision (D) coarse expression (E) deceptive appearance
12. BELEAGUER: (A) shield from (B) impress upon (C) reject (D) convert (E) discriminate
13. ABROGATE: (A) indicate (B) respond (C) review (D) violate (E) establish
14. DISCURSIVE: (A) concise (B) printed (C) profound (D) polite (E) laudatory
15. ENGENDER: (A) pervade (B) preoccupy (C) revise (D) acclimate (E) squelch

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. Until the nature of the aging process is better understood, the possibility of discovering a medicine that can block the fundamental process of aging seems very ---.

(A) urgent (B) remote (C) practical (D) substantial (E) desirable

17. Among the mysteries of the Mayan civilization are its hieroglyphics, which can only be partially ----.

(A) amended (B) obscured (C) dismantled (D) deciphered (E) improved

18. Queen Margrethe II is in fact the first formally --- Queen of Denmark named Margrethe; her ---, who ruled in the fourteenth century, was never actually crowned.

(A) contending. .antecedent
(B) ordering. .ancestor
(C) reigning. .predecessor
(D) challenging. .forerunner
(E) delegating. .introduction

19. Bees and other social insects have extremely precise mechanisms of communication, but this precision probably --- their ability to change their behavior in order to meet new situations.

(A) enhances (B) increases (C) explains (D) limits (E) interrupts

20. A stroke of humor toward the end is not sufficient to balance the ---- of the piece.

(A) harmony (B) irony (C) subtlety (D) gravity (E) flippancy

GO ON TO THE NEXT PAGE

Copyright © 1984 by Educational Testing Service. All rights reserved.

Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

In that year when the moon ate the sun, Nah-goey persuaded twenty Kogui warriors to separate temporarily from their main tribe, the Kiowas, and move southward. Two Comanches also accepted Nah-goey's pipe—An-zah-te and Tau-kan-ta-le, or Antelope Boy, who was a medicine man. It would be a difficult expedition. The men would have to cover much territory during the short hours of daylight. It was essential that the leader be resourceful and acquainted with the water holes.

The group first swam their horses across the Red River, ferrying their equipment on pontoons made of rawhide and brush. One of the ten sacred medicines of the Kiowas became waterlogged and everyone except Nah-goey, an atheist, considered this a bad sign. They hastened to build a sweat lodge in which to offer prayers of propitiation to the gods. During the ceremony, a robe thrown over the frame of the lodge caught fire. This was a warning to abandon the expedition, but Nah-goey expressed scorn for his companions' fears.

Their misgivings increased later when they frightened a hibernating bear. It stood on its hind legs, appearing to wave the warriors back. This should have been enough for anyone, but as usual the person who took a positive stand prevailed, and Nah-goey convinced the men to continue. He further discounted such omens as a flight of crows and a swarm of flies (unexplainable in this arid land).

Finally, Nah-goey said, "Who will go ahead to build an altar?" By this he was calling for an advance scout. An-zah-te offered his services. Antelope Boy acted as rear scout. It was Kiowa custom that unrecognized members of the party take the most arduous assignments. The scouts had a hard time since they had to travel farther and faster than the main body. When the two scouts returned, An-zah-te told Nah-goey of a few riders ahead, but Nah-goey remarked, "People are often fooled by herds of elk which look like men on horseback." An-zah-te was indignant at this criticism, especially as it made him a laughingstock among the men. Then Antelope Boy reported sighting a Wichita war party to the rear. He said, "Perhaps elks wear feathers in their hair and carry shields too."

But like others of various times and peoples, Nah-goey was a leader who encouraged his followers by a show of optimism, allowing it to neutralize a caution that might have saved them all. My grandfather was An-zah-te, who told me how the Kiowas mourned for Nah-goey and all his men, for they were brave Kiowa warriors.

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

21. According to the author, the position of scout was a difficult one because scouts

(A) usually bore the brunt of enemy attack
(B) had to cover more territory than the other warriors
(C) were often blamed for the failure of an expedition
(D) had to travel light, doing without certain provisions
(E) got very little gratitude or reward for their services

22. It can be inferred that, by saying the warriors "accepted Nah-goey's pipe," the author means that they

(A) drove a hard bargain
(B) planned an expedition
(C) made peace with the Kiowas
(D) agreed to follow Nah-goey
(E) offered a prayer

23. It can be inferred that the events of the story most likely took place during

(A) the dry season in the mountains
(B) the autumn after the annual flood of the Red River
(C) an unusually mild winter on the plains
(D) a season of hot days and cold, starless nights
(E) the winter of an eclipse of the sun

24. According to the author, An-zah-te, who acted as advance scout for the party, was

(A) not a Kiowa
(B) eager for glory
(C) afraid of Nah-goey
(D) familiar with the terrain
(E) the most experienced warrior

25. Which of the following would the author most likely consider to be Nah-goey's chief virtue and chief vice?

(A) Strength, opportunism
(B) Generosity, gullibility
(C) Bravery, rashness
(D) Ambition, cruelty
(E) Resourcefulness, envy

GO ON TO THE NEXT PAGE

In the eighteenth and nineteenth centuries, Newton's mechanics not only came to be thought of as compatible with common sense but had even been identified with common-sense judgment. As a result, in twentieth-century physics, the theory of relativity and the quantum theory were regarded by many as incompatible with common sense. These theories were regarded as "absurd" or, at least, "unnatural" and were rejected as Francis Bacon had rejected the Copernican system. Looking at the historical record, we notice that the requirement of compatibility with common sense and the rejection of "unnatural theories" have been advocated with a highly emotional undertone, and it is reasonable to raise the question: What was the source of heat in those fights against new and "absurd" theories? Surveying these battles, we easily find one common feature: the apprehension that a disagreement with common sense may deprive scientific theories of their value as incentives for desirable human behavior. In other words, by becoming incompatible with common sense, scientific theories lose their fitness to support desirable attitudes in the domain of ethics, politics, and religion.

Examples are abundant from all periods of theory-building. According to an old theory that was prevalent in ancient Greece and was accepted by such thinkers as Plato and Aristotle, the sun, planets, and other celestial bodies were made of a material that was completely different from the materials of which our earth consists. There were those (for example, the followers of Epicurus) who rejected this view and assumed that all bodies in the universe—earth and stars—consist of the same material. Nevertheless, many educators and political leaders were afraid that denial of the exceptional status of the celestial bodies in physical science would make it more difficult to teach the belief in the existence of spiritual beings as distinct from material things; and since it was their general conviction that the belief in spiritual beings is a powerful instrument to bring about a desirable conduct among citizens, a physical theory that supported this belief seemed to be highly desirable.

Plato, in his philosophical treatise *Laws*, suggested that people in his ideal state who taught the "materialistic" doctrine about the constitution of sun and stars would be jailed. We learn from this ancient example how scientific theories have served as instruments of indoctrination. Obviously, fitness to support the desirable conduct of citizens, or, briefly, to support moral behavior, has served through the ages as a reason for acceptance of a theory. When the "scientific criteria" did not uniquely determine a theory, its fitness to support moral or political indoctrination became an important factor for its acceptance.

26. According to the passage, heated scientific debates frequently began because

(A) a truly absurd theory had been proposed
(B) a new theory jeopardized existing moral attitudes
(C) scientists were unconcerned with politics and ethics
(D) scientific theories had been adopted too hastily
(E) a prevailing theory was incorrect but popular

27. Which of the following statements about Plato is supported by the passage?

(A) He was the originator of the theory that the material of the celestial bodies differs from that of the earth.
(B) He was a proponent of the "materialistic" theory regarding the composition of the earth and stars.
(C) He was a lawmaker who punished offenders against society.
(D) He was an advocate of the censorship of ideas and theories.
(E) He was a political leader who founded and directed an ideal state.

28. The main point of the passage is that scientific theories

(A) are sometimes misinterpreted
(B) ought not to be related to morality
(C) have to be compatible with common sense
(D) have changed through the ages
(E) have often been used to support moral rules

29. Which of the following would be the most appropriate title for the passage?

(A) The Case for Common Sense in Theory-Building
(B) A Survey of "Unnatural" Scientific Theories
(C) Factors in the Acceptance of Scientific Theories
(D) The Moral Responsibility of Scientific Investigation
(E) The Legacy of Greek Scientific Thought

30. The author's assertion that "We learn from this ancient example how scientific theories have served as instruments of indoctrination" (lines 48-50) is most weakened by the counterargument that

(A) Plato was not describing a real event and his ideal state did not exist
(B) ancient Greek ideas about matter were not really scientific theories
(C) Plato believed that the "materialistic" doctrine would lead to immorality
(D) the passage implies that scientific theories ought to be subordinate to the political needs of society
(E) the passage implies that without indoctrination there would be no stable government

GO ON TO THE NEXT PAGE

Select the word or set of words that best completes each of the following sentences.

31. Chronic criticism may be ---- habit rather than a malicious weapon, but its effects on the ---- are destructive and embittering.

(A) a thoughtless. .victim
(B) a harmless. .bystander
(C) a useless. .enemy
(D) an uncouth. .indifferent
(E) an unforgivable. .onlooker

32. Some extremist factions ---- the democratic process by denying the open competition of ---- views.

(A) repudiate. .diverse
(B) explain. .conflicting
(C) exemplify. .opposing
(D) expand. .informed
(E) subvert. .synonymous

33. What people believe about life, though it is seldom ---- their material circumstances, is by no means necessarily ---- these circumstances.

(A) reflected in. .responsive to
(B) unaffected by. .determined by
(C) applied to. .contained in
(D) dependent upon. .subordinate to
(E) implicit in. .excluded by

34. The fact that some consumer advocate committees are funded directly by the very agencies whose policies they might oppose seems to ---- the ---- of the committees.

(A) nullify. .bias
(B) preclude. .impartiality
(C) bolster. .versatility
(D) compromise. .futility
(E) assure. .objectivity

35. After the initial shock of its ---, the counterculture movement had, in the end, a rejuvenating influence on American culture in general.

(A) authoritarianism (B) iconoclasm
(C) diversion (D) speculation
(E) inspiration

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. TISSUE : HANDKERCHIEF ::
(A) cream : milk
(B) story : magazine illustration
(C) warranty : guarantee
(D) nostril : nose
(E) paper plate : dish

37. ATMOSPHERE : METEOROLOGIST ::
(A) book : botanist (B) muscle : physicist
(C) rock : geologist (D) rain : astrologist
(E) patient : pharmacist

38. ALCOVE : ROOM :: (A) niche : statue
(B) cranny : nook (C) blister : skin
(D) inlet : ocean (E) mainland : peninsula

39. AFTERTHOUGHT : IDEA :: (A) postscript : letter
(B) reflection : mirror (C) departure : journey
(D) crescendo : music (E) ingredient : recipe

40. BEQUEATH : LEGACY :: (A) achieve : goal
(B) worship : idol (C) enforce : law
(D) endow : gift (E) endure : pain

41. LABORED : SPEECH :: (A) censored : letter
(B) undisciplined : behavior (C) completed : task
(D) retraced : step (E) tortured : thought

42. SATURATE : SPRINKLE :: (A) release : free
(B) stagnate : run (C) stand : dance
(D) separate : chasten (E) vilify : admonish

43. GLUTTON : RESTRAINT ::
(A) pariah : popularity
(B) coquette : attractiveness
(C) skeptic : sarcasm
(D) perfectionist : morals
(E) enthusiast : vigor

44. PRODIGAL : SPEND :: (A) ostentatious : buy
(B) innocent : protest (C) prolific : produce
(D) dangerous : risk (E) meticulous : choose

45. BRIDLE : HORSE :: (A) bone : dog
(B) olive branch : dove (C) valor : soldier
(D) precept : conduct (E) devotion : duty

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 2 Time—30 minutes, 25 Questions

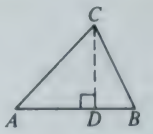
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definition of symbols:

$=$ is equal to	$\leqq$ is less than or equal to
$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. If $x + 2x + 3x = 1 + 2 + 3$, then $x =$

   (A) -1 (B) 0 (C) $\frac{1}{6}$ (D) 1 (E) 6

2. On a mathematics test, Anita solved correctly 36 of the 40 problems. What per cent of the problems did she solve correctly?

   (A) 9% (B) 10% (C) 76%
   (D) 90% (E) 96%

3. If $4^a = 16$, then $2 \times 2^a =$

   (A) 4 (B) 8 (C) 16 (D) 32 (E) 64

$$\begin{array}{r} RS \\ \times V7 \\ \hline 245 \\ \hline 245 \\ \hline 2{,}695 \end{array}$$

4. If, in the correctly computed multiplication problem above, R, S, and V each represents one of the digits 0 through 9, then $R =$

   (A) 2
   (B) 3
   (C) 4
   (D) 5
   (E) 7

5. If Mr. Smith deposits \$50 in his savings account, the total in the account will amount to twice his original deposit of \$120. What is the amount in his savings account just before this \$50 deposit?

   (A) \$10
   (B) \$70
   (C) \$170
   (D) \$190
   (E) \$290

6. If a and b are negative numbers, which of the following must be negative?

   (A) ab
   (B) $(ab)^2$
   (C) $a + b$
   (D) $a - b$
   (E) $\frac{a}{b}$

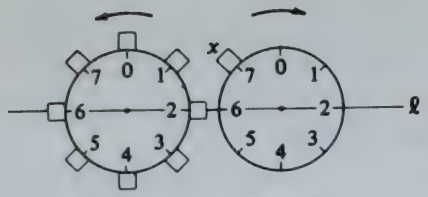
7. The gross (loaded) allowable weight of an airplane is 2,400 pounds. Its empty weight is 1,230 pounds. If the average weight per passenger of its four passengers is 150 pounds, how many additional pounds are allowable?

   (A) 570
   (B) 600
   (C) 1,020
   (D) 1,800
   (E) 2,250

8. If $x \neq 0$, $x = y$, and $\frac{3a}{x} = \frac{9b}{y}$, what is the value of a in terms of b?

   (A) $\frac{b}{3}$ (B) b (C) $3b$ (D) $6b$ (E) $9b$

GO ON TO THE NEXT PAGE



9. In the figure above, wheel B moves in a clockwise direction. Whenever cog x meets wheel A along line ℓ, it causes wheel A to move from one cog to the next in a counterclockwise direction. How many complete revolutions of wheel B are necessary to move wheel A 2 complete revolutions?

(A) 2 (B) 4 (C) 8 (D) 12 (E) 16

---

10. The product of 3, 4, and 5 is equal to twice the sum of 10 and

(A) 3
(B) 20
(C) 30
(D) 50
(E) 110

11. If the area of each of the following rectangles is 1, which has a length of $\frac{3}{2}$?

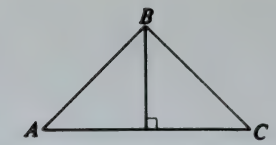
(A) Length $\frac{2}{3}$

(B) Length $\frac{1}{2}$

(C) Length $\frac{3}{4}$

(D) Length $\frac{3}{2}$

(E) Length $\frac{1}{3}$



12. In $\triangle ABC$ above, if $AB = BC$ and if $\angle ABC$ is a right angle, exactly how many 45-degree angles are formed by pairs of the line segments?

(A) None (B) Two (C) Three
(D) Four (E) Five

TRISCHOOL TRACK MEET

	Event I	Event II	Event III
First Place (5 Points)			
Second Place (3 Points)			
Third Place (1 Point)	A		A

13. The figure above is a partially completed score card for a trischool track meet. Schools A, B, and C each entered one person in each of the three events. Each school ran in every event and there were no ties. What is the LEAST possible score for School B at this meet?

(A) 1
(B) 7
(C) 9
(D) 10
(E) It cannot be determined from the information given.

GO ON TO THE NEXT PAGE

14. To make fruit punch for 8 people, $\frac{1}{2}$ liter of concentrate is mixed with $3\frac{1}{2}$ liters of water. How many liters of this concentrate are needed to make fruit punch of this strength for 28 people?

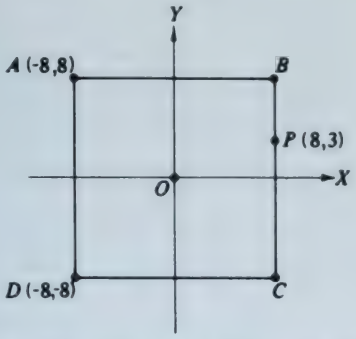
(A) 1

(B) $1\frac{3}{4}$

(C) 2

(D) $3\frac{1}{2}$

(E) 7



15. In the figure above, if Q is a point (not shown) such that segment PQ bisects the area of the square region $ABCD$, what are the coordinates of Q?

(A) $(-8, -3)$ (B) $(-8, 3)$ (C) $(3, -8)$
(D) $(8, -3)$ (E) $(3, 8)$

16. In the land of Rad, the Holiday of the Harvest Moon is always celebrated on the third Monday in September. The earliest day in September that the holiday could occur is

(A) Sept. 14 (B) Sept. 15 (C) Sept. 20
(D) Sept. 21 (E) Sept. 22

17. If $\frac{x}{2} = y$ and $2y = y$, what is the value of x?

(A) 0 (B) 1 (C) 2 (D) 4
(E) It cannot be determined from the information given.

18. Of the following, which is closest to 5?

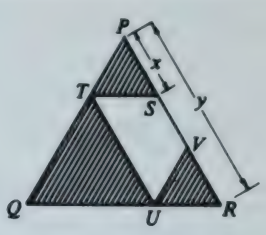
(A) $\frac{5.17 \times 4.91}{0.51}$

(B) $\frac{51.7 \times 4.91}{0.51}$

(C) $\frac{51.7 \times 4.91}{5.1}$

(D) $\frac{0.517 \times 49.1}{5.1}$

(E) $\frac{51.7 \times 49.1}{51}$



19. In the figure above, ΔPQR and the three shaded triangles are equilateral. If $PR = y$ and $PS = x$, what is the perimeter of quadrilateral $STUV$ in terms of x and y?

(A) $2y$ (B) $3y - 2x$ (C) $2y - x$
(D) $2y + x$ (E) $2y - 2x$

20. This question was not counted in computing scores.

GO ON TO THE NEXT PAGE

21. Points P, Q, and R lie on a straight line. P is 3 kilometers from R, which is 2 kilometers from Q. What is the distance, in kilometers, between P and Q?

(A) 1
(B) 3
(C) $\sqrt{13}$
(D) 5
(E) It cannot be determined from the information given.

22. For all integers x and y, let $x \star y = 3x + 2y$. Which of the following must be true?

I. $3 \star 2 = 13$
II. $(0 \star 1) \star 2 = 0 \star (1 \star 2)$
III. $x \star y = y \star x$

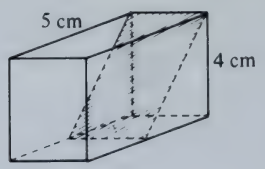
(A) I only (B) II only (C) III only
(D) II and III only (E) I, II, and III

23. A wheel with radius 0.75 meters covers 1,000 meters in one hour without slipping. How many revolutions per minute does the wheel make?

(A) $\frac{4{,}000\pi}{3}$
(B) $\frac{4{,}000}{3\pi}$
(C) $\frac{2{,}000}{3\pi}$
(D) $\frac{200\pi}{9}$
(E) $\frac{100}{9\pi}$

24. Which of the following indicates the set of all numbers x such that there is a triangle with sides of lengths $x, x-2$, and $7-x$?

(A) $2 < x$
(B) $x < 7$
(C) $0 < x < 9$
(D) $2 < x < 5$
(E) $3 < x < 5$



25. In the figure above, the empty box has a flap lid of area 15 cm^2. One of the three unattached edges of the lid rests on the bottom of the box as shown and separates the box into two compartments. If it is assumed that the lid is rigid and fits tightly, the larger, open compartment has volume

(A) 18 cm^3 (B) 24 cm^3 (C) 36 cm^3
(D) 42 cm^3 (E) 60 cm^3

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 4 Time—30 minutes 40 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. UNCOMMON: (A) small and unimportant (B) frequent and usual (C) monotonous (D) predictable (E) exclusive

2. DISSOCIATE: (A) reveal (B) cleanse (C) join together (D) increase rapidly (E) approve of

3. VULGARITY: (A) honesty (B) strong conviction (C) marked similarity (D) refinement (E) binding declaration

4. CONTEMPT: (A) respect (B) ability (C) confusion (D) discretion (E) revelation

5. COMPATIBILITY: (A) lack of freedom (B) lack of accord (C) need for expansion (D) need for training (E) need for change

6. EFFERVESCE: (A) display boredom (B) suffer injury (C) endure hardship (D) become definite (E) speak frankly

7. URBANE: (A) greedy (B) blameworthy (C) powerful (D) delightful (E) boorish

8. LARGESS: (A) solemnity (B) discontent (C) inferiority (D) imperfection (E) miserliness

9. DIURNAL: (A) occurring on land (B) occurring frequently (C) occurring at night (D) occurring suddenly (E) occurring in secret

10. APOCRYPHAL: (A) irrevocable (B) immeasurable (C) distinguishable (D) of significant value (E) of known origin

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Although she enjoys her fame, Andrienne is a woman who craves ---, who needs quiet and repose.

(A) suspense (B) ceremony (C) solitude (D) ferment (E) prosperity

12. Much of ancient Africa became progressively ---, with forest giving way to savanna, savanna to prairie, and prairie to ---.

(A) populated. .jungle
(B) domesticated. .wildlife
(C) savage. .farmland
(D) desiccated. .desert
(E) polluted. .cultivation

13. The --- of the instrument proved the --- care with which it had been constructed.

(A) precision. .meticulous
(B) usefulness. .scanty
(C) defects. .painstaking
(D) inefficiency. .artistic
(E) expensiveness. .scientific

14. Lakes tend to become --- with time because their waves --- promontories that jut out into them.

(A) round. .wear away (B) dangerous. .avoid (C) longer. .wash over (D) irregular. .demolish (E) shallow. .break against

15. An allusion is highly --; like a richly connotative word or symbol, it suggests far more than it says.

(A) simplistic (B) evocative (C) descriptive (D) imaginary (E) persuasive

GO ON TO THE NEXT PAGE

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. CRUMB : BREAD :: (A) ounce : unit
(B) splinter : wood (C) water : bucket
(D) twine : rope (E) bit : fraction

17. TELESCOPE : MAGNIFICATION ::
(A) architecture : acoustics
(B) treaty : ratification
(C) umbrella : rain
(D) megaphone : amplification
(E) microscope : microorganism

18. UNMASK : DISGUISE :: (A) introduce : speaker
(B) inscribe : message (C) dissolve : water
(D) ruffle : hair (E) disrobe : garment

19. DRAFT : NOVELIST :: (A) stanza : poet
(B) diagram : grammarian (C) fabric : seamstress
(D) sketch : painter (E) lens : photographer

20. EMBARK : JOURNEY :: (A) capture : prey
(B) copy : invention (C) commence : activity
(D) interrupt : procedure (E) extinguish : fire

21. DROWSY : SLEEP :: (A) weary : strive
(B) curious : hide (C) sluggish : fidget
(D) cunning : blunder (E) edgy : fret

22. ACCELERATE : VELOCITY ::
(A) deplete : supply (B) surround : width
(C) enlarge : size (D) hurry : distance
(E) evaluate : worth

23. ANARCHIC : GOVERNMENT ::
(A) freethinking : dogma
(B) malfunctioning : eclipse
(C) tyrannical : monarchy
(D) plagiarized : scholarship
(E) stern : discipline

24. CELEBRITY : ACCLAIM ::
(A) hero : battle (B) leader : deference
(C) outcast : flattery (D) admirer : limelight
(E) pioneer : immigration

25. TURPITUDE : SCOUNDREL ::
(A) impassivity : scholar
(B) initiative : conformist
(C) insignificance : nonentity
(D) incompetence : expert
(E) impotence : follower

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

By 1746, England had, once and for all, turned civilian. Afterward the English fought many wars, but they took place a long way off, and the contestants were regular soldiers fighting for pay. People seem to have felt about the sword in those days as Dr. Johnson said he thought about the pen when he declared that no one but a blockhead ever wrote except for money. The age of economists and calculators had come. Whether it was Napoleon or Adam Smith who called the English a nation of shopkeepers, it was someone speaking in the eighteenth century, the age of Jeremy Bentham and his gospel of utility. In his *Fragment on Government* in 1776, that economist and philosopher begged his readers not to blame him if he wrote "a mercenary language"; if we are not to say that happiness or pleasure can be measured in terms of money, is there anything meaningful that the legislator can say about it, he wondered. Quantity, not quality, alone is measurable, and law and the legislator can be, and should be, interested in nothing else.

This is one of the things that repelled, and continues to repel, many people about eighteenth-century England. "Soul dead, stomach well-alive" was how Thomas Carlyle was to summarize a century that ended with Bentham's "Felicific Calculus," a calculus of pleasures and pains.

26. The passage as a whole is concerned with

(A) the financial status of people in eighteenth-century England
(B) a marked change in attitude of the common people toward the intrinsic value of work
(C) the search for tangible rewards for what was once voluntary labor
(D) the development of a society that emphasized business and measured worth in terms of economic gain
(E) economic development that no longer emphasized class distinctions but found that each person contributed to financial growth

27. In the context of the passage, the phrase "a nation of shopkeepers" (lines 10-11) is probably meant as

(A) a compliment, because it recognizes the business skill of the enemy
(B) a denunciation, because it suggests the abusive treatment of the impoverished in that age
(C) a call to arms, for it suggests the prejudices that the English needed to overcome
(D) a warning, for it analyzes the intelligence and even the craftiness that brought English business such huge profits
(E) an insult, for it suggests the narrow interests and profit-centered motives of the English of that time

28. According to the passage, Bentham's philosophy is basically an attempt to

(A) suggest the inadequacy of then current lifestyles because they relied to a great extent on finance
(B) emphasize the value of happiness, which is not easily measured
(C) reject pleasure as a possibility in the human sphere of activity because it cannot be measured
(D) suggest that what is valuable in life can be quantified
(E) invalidate the assumed quality of eighteenth-century life

GO ON TO THE NEXT PAGE

Bacteria stick, often with exquisite specificity, to surfaces ranging from the human tooth or lung to a rock submerged in a stream. They do so by means of a mass of tangled fibers of polysaccharides, or branching sugar molecules, that extend from the bacterial surface and form a feltlike glycocalyx surrounding an individual cell or a colony of cells. The adhesion mediated by the glycocalyx explains particular locations of bacteria in most natural environments; more specifically, it is a major determinant in the initiation of bacterial diseases ranging from dental caries to pneumonia.

These major—and with the benefit of hindsight, obvious—facts about the bacterial cell surface have become known only within the past decade. Ironically, the main reason for the late discovery of the bacterial glycocalyx was the long reliance of microbiologists on an otherwise eminently effective investigative system: the pure laboratory culture of an individual bacterial strain. To generate and maintain a glycocalyx, a bacterial cell must expend energy, and in the protected environment of a pure culture, the glycocalyx is a metabolically expensive luxury conferring no selective advantage. Cells that fabricate these elaborate coatings are usually eliminated from pure cultures by uncoated mutants that can devote more of their energy budget to proliferation. It is these uncoated mutants that microbiologists have usually studied.

29. Which of the following can be inferred from the passage about bacteria that expend energy to produce adhesive capacity in a laboratory culture?
    (A) They have a limited advantage over those that do not.
    (B) They are likely to proliferate more rapidly than uncoated bacteria.
    (C) They waste resources that would enhance their chances for survival.
    (D) They draw energy away from neighboring uncoated bacteria.
    (E) They are more resistant to disease than are uncoated bacteria.

30. The passage indicates that the failure of microbiologists to recognize bacterial adhesive properties is best described as
    (A) an inevitable result of studying laboratory cultures
    (B) a result of careless laboratory techniques
    (C) an astounding lack of scientific judgment
    (D) a confused reaction to previous overemphasis on bacterial mutations
    (E) a result of an inability to identify all the surfaces adhered to by bacteria

31. Which of the following statements concerning the specificity of bacterial adhesion is (are) supported by the passage?
    I. It is indispensable in bacterial proliferation.
    II. It is less evident in laboratory cultures.
    III. It is unimportant in most natural environments.
    (A) I only (B) II only (C) III only
    (D) I and II only (E) II and III only

32. The primary purpose of the passage is to
    (A) explain a previous gap in the knowledge of bacterial surface function
    (B) describe the various surfaces to which bacteria can adhere
    (C) call for reevaluation of current microbiological beliefs
    (D) demonstrate how bacterial disease can be initiated by adhesion
    (E) explore the possibility of making environmental bacteria nonadhesive

GO ON TO THE NEXT PAGE

Many social anthropologists and other scientific observers of human communities have emphasized the similarities in the sex roles in various communities. One very distinguished anthropologist, Margaret Mead, in her book *Male and Female*, gives this summary description of the sex roles: "The home shared by a man or men and female partners, into which men bring the food and women prepare it, is the basic common picture the world over. But this picture can be modified, and the modifications provide proof that the pattern itself is not something deeply biological."

It is surprising that Margaret Mead, with her extensive and intensive personal experience of diverse communities throughout the world, should venture upon such a dubious generalization. She is right in describing the preparation of food as a monopoly for women in nearly all communities, but the surmise that the provision of food is a man's prerogative is unwarranted. In fact, an important distinction can be made between two kinds or patterns of subsistence agriculture: one in which food production is taken care of by women, with little help from men, and one in which food is produced by the men with relatively little help from women. As a convenient terminology I propose to denote these two systems as the female and male systems of farming.

33. Which of the following best explains what the author means by "the similarities in the sex roles" (lines 2-3) ?

(A) The equality of men's and women's traditional tasks
(B) The likenesses in patterns of division of labor between men and women
(C) The universal acceptance of the need for co-operation between men and women within a community
(D) The overlapping of tasks performed by men and women in various communities
(E) The correspondence between a community's attitude toward women and the traditional tasks they perform

34. The author's attitude toward the statement by Margaret Mead is one of

(A) reluctant consent
(B) intrigued curiosity
(C) respectful disagreement
(D) apologetic defensiveness
(E) mild endorsement

35. Which of the following best describes the relation between the two paragraphs in the passage?

(A) The second disputes aspects of the opinions presented in the first.
(B) The second explains the logic behind the arguments summarized in the first.
(C) The second provides specific examples of the general statements presented in the first.
(D) The second questions the social importance of the issues raised in the first.
(E) The second analyzes the implications for the future of the theories described in the first.

GO ON TO THE NEXT PAGE

Areopagitica, written in 1643 and published in 1644, is one of Milton's important prose works. The ordinance of Parliament that occasioned it—an act requiring that all books be licensed by an official censor before publication—had been passed in June 1643. Milton addressed the work to Parliament in the hope of influencing its members to repeal a decree inconsistent, in his view, with their own history and purposes as the restorers of English liberties. After complimenting this body on its past achievements and excusing himself for venturing this criticism of their recent act, Milton reviews the history of licensing from ancient times, showing that it has always been a concomitant of tyranny. Secondly, he enters upon a noble defense of the benefit of books freely used, showing how necessary reading of every sort is to the attainment of knowledge and experience in a world where good and evil grow up indiscriminately together. Next, he deals with the impossibility of the attempt to make people virtuous by external restraint. Corrupting influences are present everywhere and can be met only by building up inner discipline and the power of rational choosing. This is the fundamental tenet of Milton's ethical philosophy. He goes on to argue that the present law will be ineffective even as applied to publication. *Areopagitica* itself was published in defiance of the law. Finally, Milton attacks the order as a discouragement to intellectual activity and a hindrance to the cause of truth. The plea here is one of resolute faith in the competence of human nature, and particularly of English nature, to work out its own intellectual salvation, and in the divine property of Truth itself, which is sure to prevail ultimately over error if the two are allowed to grapple. Milton expresses with extraordinary forcefulness the resentment of the mature mind at being kept under watch and ward, and vividly displays the results of such policy in spiritual stagnation and a "starched conformity of opinion." The spirit of English protest speaks through him; his principles are as vital and his warnings as necessary today as they ever were.

Milton's plea for liberty failed in its immediate purpose, but it was employed when the issue was raised again in 1679 and 1693; it furnished Mirabeau with the substance of a pamphlet on the freedom of the press in 1788; and it has armed the minds of individuals in all times against the ever-recurring attempt to silence thought.

36. The primary emphasis of the passage is on

(A) presenting the historical background of the publication of *Areopagitica*
(B) detailing the influence of Milton's ethical philosophy
(C) arguing against Parliament's system of censorship
(D) sketching an overview of the arguments contained in *Areopagitica*
(E) debating the validity of Milton's ethical philosophy in *Areopagitica*

37. According to the passage, Milton believed that the licensing of books would result in which of the following?

I. The spread of corruption
II. The hindrance of truth
III. Spiritual stagnation

(A) II only (B) III only (C) I and II only (D) I and III only (E) I, II, and III

38. The author's tone in the second paragraph can best be described as

(A) carefully impartial (B) frankly approving (C) faintly critical (D) notably ambivalent (E) thoroughly disapproving

39. It can be inferred from the passage that Milton believed that Parliament's moral responsibility to the English public was to

(A) lead by its good example
(B) control major corrupting influences
(C) dictate public morality through noncoercive means
(D) punish only individuals who defied the law
(E) allow the public full freedom in moral matters

40. It can be argued that the ideas contained in *Areopagitica* are most applicable today in defense of

(A) judicious censorship
(B) copyright legislation
(C) freedom of religion
(D) freedom of the press
(E) democratic elections

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

SECTION 5 Time—30 minutes 35 Questions

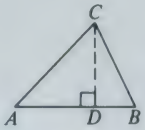
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definition of symbols:

$=$ is equal to	$\leq$ is less than or equal to
$\neq$ is unequal to	$\geq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. $\dfrac{3^2+3^2+3^2}{2^3+2^3+2^3} =$

(A) $\frac{9}{8}$ (B) 1 (C) $\frac{8}{9}$ (D) $\frac{3}{4}$ (E) $\frac{729}{1{,}028}$

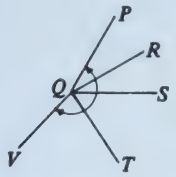
2. Which of the following products is (are) equal to 1 ?

I. $\frac{2}{3} \times \frac{3}{2} \times \frac{2}{3} \times \frac{3}{2}$

II. $\frac{2}{3} \times \frac{3}{2} \times \frac{2}{3} \times \frac{3}{2} \times \frac{2}{3}$

III. $\frac{2}{3} \times \frac{3}{2} \times \frac{2}{3} \times \frac{3}{2} \times \frac{2}{3} \times \frac{3}{2}$

(A) I only
(B) II only
(C) I and III only
(D) II and III only
(E) I, II, and III



Note: Figure not drawn to scale.

3. In the figure above, if $\angle PQS = 60°$, $\angle RQT = 70°$, $\angle TQV = 50°$, and $\angle RQS = 30°$, then the marked $\angle PQV =$

(A) 80° (B) 120° (C) 150°
(D) 210° (E) 220°

4. If $S = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5}$ and

$T = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$, then $S - T =$

(A) $-\frac{1}{30}$ (B) $-\frac{1}{5}$ (C) $\frac{1}{30}$ (D) $\frac{1}{5}$ (E) $\frac{131}{30}$

5. If the average of $-12, -5, -3, -6$, and y is -6, then $y =$

(A) -56
(B) -4
(C) 2
(D) 4
(E) 14

6. If $\boxed{x,y}$ is defined as $\dfrac{x-1}{y+1}$ where x and y are positive integers, which of the following is least?

(A) $\boxed{2,2}$ (B) $\boxed{3,3}$ (C) $\boxed{5,3}$
(D) $\boxed{3,5}$ (E) $\boxed{2,4}$

7. The length of a certain rectangle is 3 times its width. If x represents the width of the rectangle, which of the following represents the perimeter of the rectangle?

(A) $4x$
(B) $\frac{8}{3}x$
(C) $8x$
(D) $3x + 2$
(E) $3x^2$

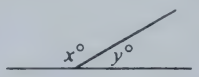
GO ON TO THE NEXT PAGE

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet fill in oval

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

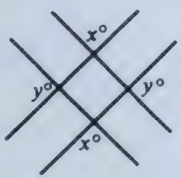
AN E RESPONSE WILL NOT BE SCORED.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ Ⓔ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ Ⓔ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ● Ⓔ

Notes:

1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

	Column A	Column B
8.	0.21	0.0392
	Scott is taller than Kathy and Maria is taller than Kathy.	
9.	Scott's height	Maria's height
	$\sqrt{25} + \sqrt{144} = x$	
10.	13	x
11.	$3\left(\frac{1}{2} + \frac{1}{8}\right)$	$\frac{3}{2} + \frac{3}{8}$
	$\frac{3}{5} = \frac{x}{20}$ $\frac{4}{8} = \frac{y}{24}$	
12.	x	y
13.	 $x + y$	170

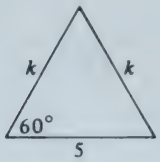
	Column A	Column B
14.	$10 \cdot 9 \cdot 8 \cdot 7$	$70 \cdot 72$
	m is an even integer and n is an odd integer.	
15.	The probability that $m + n$ is even	The probability that mn is even
16.	$1^{10} + 2^{10}$	3^{10}
17.	The number of positive integer factors of 7	The number of positive integer factors of 19
18.	$\frac{10}{11}$	$\frac{11}{12}$
	$x + y + z = 8$ and $y = z$	
19.	z	4

GO ON TO THE NEXT PAGE

SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

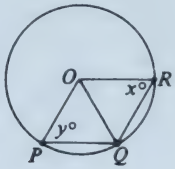
Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

AN E RESPONSE WILL NOT BE SCORED.

	Column A	Column B
20.	 $2k$	10
21.	The total surface area of a cube having edge 0.5	2

$y = (x - 1)(x + 2)$

22.	The least value of x for which $y = 0$	-1



O is the center of the circle and segment $PQ >$ segment QR.

23.	x	y

	Column A	Column B
24.	Area of a square with a diagonal of length 4	Area of a parallelogram with a diagonal of length 4

On a certain blueprint, 1 centimeter represents 2 meters.

25.	The area represented by 29 square centimeters on this blueprint	116 square meters
26.	$\sqrt{x^4 + 4x^2 + 4}$	$x^2 + 2x + 2$

Let $\triangle$ be defined by the equation $a \triangle b = a^3$.

27.	$x \triangle 4$	$x \triangle 5$

GO ON TO THE NEXT PAGE

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

28. If P is the greatest positive 4-digit integer with nonzero digits, none of which is repeated, and Q is the least of such positive integers, then $P - Q =$

(A) 2,468
(B) 5,555
(C) 8,642
(D) 8,646
(E) 8,999

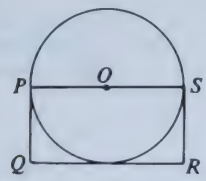


29. Which of the following designs can be formed by combining rectangles with size and shading the same as that shown above if overlap is not permitted?



I. II. III.

(A) I only (B) II only (C) I and II only
(D) II and III only (E) I, II, and III



30. In the figure above, $PQRS$ is a rectangle with sides PQ, QR, and RS touching the circle with center O. If the diameter of the circle is x, what is the perimeter of $PQRS$ in terms of x?

(A) $2x^2$ (B) $\frac{x^2}{2}$ (C) $6x$ (D) $5x$ (E) $3x$

31. A man willed $\frac{2}{5}$ of his estate to his daughter and $\frac{1}{20}$ to each of his three sons. Another $\frac{1}{10}$ was divided equally among his five grandchildren. If the remainder of the man's estate was left to his wife, what part of his estate did his wife receive?

(A) $\frac{1}{5}$ (B) $\frac{3}{10}$ (C) $\frac{7}{20}$ (D) $\frac{2}{5}$ (E) $\frac{9}{20}$

32. If car X starts from point P and travels at an average rate of 50 kilometers per hour and car Y starts from point P 6 hours later and travels on the same route at an average rate of 60 kilometers per hour, exactly how many hours after car Y leaves P will it overtake car X?

(A) 36 (B) 30 (C) 24 (D) 11 (E) 5

33. If $x \neq 0$ and $x = 3y = 5t$, what is the value of $x - y$ in terms of t?

(A) $\frac{1}{5}t$

(B) $\frac{5}{3}t$

(C) $2t$

(D) $\frac{10}{3}t$

(E) $10t$

34. On a biology quiz, the average score for a class was 80. If 20 per cent of the class scored 90 and 30 per cent scored 70, what was the average score for the remainder of the class?

(A) 76 (B) 78 (C) 80 (D) 82 (E) 84



35. If the shaded region above is bounded by three nonoverlapping semicircles each with a radius of 1, what is the area of the region?

(A) $\frac{\sqrt{3}}{4} + \frac{3\pi}{8}$

(B) $\sqrt{3} + \frac{3}{2}\pi$

(C) $2 + \frac{3}{2}\pi$

(D) 3π

(E) $\frac{\sqrt{2}}{2} + 3\pi$

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

Correct Answers for Scholastic Aptitude Test
Form Code 4W

VERBAL		MATHEMATICAL	
Section 1	Section 4	Section 2	Section 5
1. E	1. B	1. D	1. A
2. B	2. C	2. D	2. C
3. C	3. D	3. B	3. C
4. C	4. A	4. B	4. D
5. D	5. B	5. D	5. B
6. E	6. A	6. C	6. E
7. B	7. E	7. A	7. C
8. A	8. E	8. C	*8. A
9. C	9. C	9. E	*9. D
10. E	10. E	10. B	*10. B
11. D	11. C	11. A	*11. C
12. A	12. D	12. D	*12. C
13. E	13. A	13. B	*13. A
14. A	14. A	14. B	*14. C
15. E	15. B	15. A	*15. B
16. B	16. B	16. B	*16. B
17. D	17. D	17. A	*17. C
18. C	18. E	18. D	*18. B
19. D	19. D	19. C	*19. D
20. D	20. C	20. †	*20. C
21. B	21. E	21. E	*21. B
22. D	22. C	22. A	*22. B
23. E	23. A	23. E	*23. A
24. A	24. B	24. E	*24. D
25. C	25. C	25. D	*25. C
26. B	26. D		*26. D
27. D	27. E		*27. C
28. E	28. D		28. C
29. C	29. C		29. D
30. A	30. A		30. E
31. A	31. B		31. C
32. A	32. A		32. B
33. B	33. B		33. D
34. B	34. C		34. D
35. B	35. A		35. B
36. E	36. D		
37. C	37. E		
38. D	38. B		
39. A	39. E		
40. D	40. D		
41. E			
42. E			
43. A			
44. C			
45. D			

*Indicates four-choice questions. (All of the other questions are five-choice.)
†Not scored

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.

- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

SAT-Verbal Sections 1 and 4

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 4* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 218. Enter this figure on the worksheet.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 218. Enter this figure on the worksheet.

SAT SCORING WORKSHEET

SAT-Verbal Sections

A. Section 1: _____ no. correct − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 4: _____ no. correct − ¼ (_____ no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on page 218.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 2: _____ no. correct − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ no. correct − ¼ (_____ no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ no. correct − ⅓ (_____ no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on page 218.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Code 4W

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	470	610
84	790		39	460	600
83	780		38	450	600
82	770		37	450	590
81	760		36	440	580
80	750		35	430	570
79	740		34	430	560
78	730		33	420	550
77	720		32	410	540
76	710		31	410	530
75	700		30	400	520
74	700		29	390	510
73	690		28	390	510
72	680		27	380	500
71	670		26	370	490
70	670		25	370	480
69	660		24	360	470
68	650		23	350	460
67	650		22	350	450
66	640		21	340	440
65	630		20	330	430
64	620		19	330	420
63	620		18	320	410
62	610		17	310	410
61	600		16	310	400
60	600		15	300	390
59	590	800	14	300	380
58	580	780	13	290	370
57	580	770	12	280	360
56	570	760	11	270	350
55	570	750	10	270	340
54	560	740	9	260	340
53	550	730	8	250	330
52	550	720	7	250	320
51	540	710	6	240	310
50	530	700	5	230	300
49	530	690	4	220	290
48	520	680	3	220	280
47	510	670	2	210	280
46	510	660	1	200	270
45	500	660	0	200	260
44	490	650	−1	200	250
43	490	640	−2	200	240
42	480	630	−3	200	230
41	470	620	−4	200	230
			−5	200	220
			−6 or below	200	200